

External Tables



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What is a Snowflake External Table ?

A Snowflake External table is a Snowflake table type that allows us to query data stored in flat files on external cloud storage like GCP, AZURE and AWS(s3) using Snowsight and SnowSQL.

File data can be accessed via Snowsight which eliminates the need to load the data into Snowflake before you can access them.

Reading data from s3 instead of loading data to Snowflake tables saves on compute costs and time

File stored on cloud storage(AWS s3) are accessible to different applications like Amazon Athena, Redshift Spectrum, Amazon Sagemaker in addition to Snowflake External tables.

When datafiles in AWS s3 changes they are available to all applications to access instead of needing to load delta data into each application again

Snowflake External table data is stored outside of Snowflake and we do not get charged for storage by Snowflake.

Snowflake stores file level metadata like file names associated with the External table and related properties like row count.

What is an External Table ?

Data can be stored on cloud storage in several formats and File format supported by External tables include all the file formats that are supported by the COPY INTO command and include [CSV|JSON|AVRO|ORC|PARQUET|XML](#).

External tables are read only and we cannot perform DML operations like INSERT,UPDATE, DELETE.

External tables can suffer from performance issues if there are many files associated with it.

We can partition External tables which may help reduce performance issues.

Query response time for partitioned External tables is faster as the query can go directly to the partition that has the relevant data instead of scanning all files.

Auto-Refresh in External Tables

External tables can frequently receive changes in the files associated with it. Files could be added or deleted. When files associated with External table change the External table is not automatically aware of the files that have been added or removed. Hence data from new files that are added will not be accessible to the External table.

To be able to read data files that have been newly added the files will have to be registered with the External table and the process of registering the new files is called refresh in External tables.

Refresh can be done manually or automatically.

We can use `ALTER EXTERNAL TABLE <tablename> REFRESH` to manually refresh the external table.



For automatic refresh we can enable s3 Event notification which will generate a notification when a file is added or removed from s3. This notification is then sent to an AWS SQS queue. Snowflake External table will poll for any AWS SQS notifications and if it receives them it will now be aware of the arrival or removal of files and the Snowflake External table metadata will be updated, the new data will be available for select from the External Table.

Partitioned External Tables

External tables can be created either as partitioned or non-partitioned.



In production the AWS s3 path associated with an External table could have 1000's of files.

External table SELECT performance declines as the number of files increases as it must scan 1000's of files to locate the data requested based on the WHERE clause and files are not optimized for read performance.

How can we fix this performance challenge with External tables ?

We can partition the External table based on the most frequently used column in the WHERE clause. Once the partition is implemented the SELECT statement skips all partitions that are not mentioned in the WHERE clause and this reduces the number of files to be read.

Partitions need to be created on s3 and files should be placed inside these partitions. External table should be created in Snowflake and it should reference the partitions created on s3.

Benefit of partitions is only if we use the partition column in the where clause.

External Tables in non-csv file format.

External tables can be created in several different file format in addition to CSV

Supported file formats are PARQUET,ORC,AVRO,JSON,XML in addition to CSV.



Data Lakes store data in mainly PARQUET or ORC format due to columnar storage and lower space consumption which make it more performant when storing and reading data from an AWS Data Lake.

Benefits:

- Data stored in PARQUET/ORC format return data faster than CSV files.
- Data stored in PARQUET/ORC format due to columnar storage consume less credits for SELECT.
- Data stored in PARQUET/ORC format due to columnar storage consume less space than CSV file format and we get 87% saving in space consumption.
- Data types can be stored inside PARQUET/ORC which improves data ingestion speed.
- Data types can be stored inside PARQUET/ORC which improves data compression.
- Data stored in PARQUET works great for Spark based workloads.

JSON/XML/AVRO:

- JSON file format is extensively used in IOT device data.
- XML file format is extensively used in data exchange between applications.
- AVRO file format is used with streaming platforms like Apache Kafka and these file formats are extensively used and stored in the AWS Data Lake.

External Tables in JSON file format.

IOT(Internet of Things) device is a sensor which provides information on the health or behavior of a device.

IOT data from devices generally are in JSON format and being able to create a Snowflake External table on JSON file data simplifies data analysis and data ingestion of IOT data.

Benefits:

- JSON External tables eliminates need to use programming like python to ingest data.
- Being able to create External tables saves time and data engineer effort as data can be ingested or read without having to run the COPY command

```
{
  "COMMISSION_PCT": null,
  "DEPARTMENT_ID": 100,
  "EMAIL": "DFAVIET",
  "EMPLOYEE_ID": 109,
  "FIRST_NAME": "DANIEL",
  "HIRE_DATE": "2002-08-16",
  "JOB_ID": "FI_ACCOUNT",
  "LAST_NAME": "FAVIET",
  "MANAGER_ID": 108,
  "PHONE_NUMBER": "515.124.4169",
  "SALARY": 9000
},
{
  "COMMISSION_PCT": null,
  "DEPARTMENT_ID": 100,
  "EMAIL": "JCHEN",
  "EMPLOYEE_ID": 110,
  "FIRST_NAME": "JOHN",
  "HIRE_DATE": "2005-09-28",
  "JOB_ID": "FI_ACCOUNT",
  "LAST_NAME": "CHEN",
  "MANAGER_ID": 108,
  "PHONE_NUMBER": "515.124.4269",
  "SALARY": 8200
},
```

{JSON}

Streams on External Tables.

Snowflake supports creation of Streams on External tables.

New data files that are added to the External table path are immediately available in the Stream for ingestion into Snowflake.

Streams on External table are INSERT_ONLY which means that rows from data files that are added will appear in the Stream, however rows from any file that has been removed will not show up as delete rows.

Streams on External tables improves data latency for time-sensitive data as data is immediately available for ingestion into Snowflake which can be accomplished with the help of a Task that polls the Stream for data.



`WHEN SYSTEM$STREAM_HAS_DATA('<stream_name>')`

Credit savings as the Task would execute only if there is data in the Stream.

Stream can be created with the

```
CREATE OR REPLACE STREAM <stream_name> on EXTERNAL TABLE <tablename>  
INSERT_ONLY=TRUE;
```

Streams on External Tables.

Pre-requisite:

- s3 Event notifications must be set for the External table in order for the data to show up immediately in the Stream.
- Use `ALTER EXTERNAL TABLE <tableName> REFRESH` to manually refresh the External table would cause the stream to be populated with the rows from the new data file.

Streams on External Tables.(v2)

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```

Summary

- Snowflake External tables are designed to read data files from cloud storage (GCP, AZURE AWS).
- Snowflake External tables are read only and cannot be updated or deleted.
- Snowflake External tables can be auto refreshed by setting up event notification on cloud storage.
- Snowflake External tables can be manually refreshed by `ALTER EXTERNAL TABLE <tablename> REFRESH.`
- External table SELECT performance declines as the number of files increase, Snowflake External tables can be partitioned to improve query performance, this happens by Snowflake skipping partitions that do not have the relevant data.
- Snowflake External tables support several file formats including parquet, ORC, AVRO, JSON, XML and CSV
- External tables are created on JSON files to ingest IOT data.

Summary

- Streams can be created on External tables which enables near real time ingestion of External table data.
- Streams on External tables are INSERT_ONLY , newly inserted rows will appear in the Stream, deleted and updated rows will be ignored.
- Snowflake External tables can be used to ingest large data files over 250 mb which are not suitable for Snowpipe.
- External tables are created by using **EXTERNAL** in the create table statement and providing the stage information and File format which are mandatory and optional pattern.

Thank you



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